EXPERIMENT 24

Dialog Act Recognition in Conversational Utterances

AIM

To create a Python program that recognizes and classifies dialog acts (e.g., Greet, Question, Inform, Request, Acknowledge, Farewell) in a conversational exchange.

PROCEDURE

1. Define taxonomic dialog act categories based on communicative intent (DAMSL/Switchboard schema).
2. Establish lexical cue patterns, syntactic structures, and punctuation indicators for dialog acts.
3. Implement a classification pipeline that analyzes utterance surface forms.
4. Parse dialogue turns between interlocutors and tag each turn with its corresponding dialog act.
5. Display the dialogue sequence labeled with recognized dialog acts.

PROGRAM

```
import re

class DialogActClassifier:
    def __init__(self):
        self.rules = [
            (r'^(hi|hello|hey|good morning|greetings)\b', 'GREETING'),
            (r'^(bye|goodbye|see you|farewell)\b', 'FAREWELL'),
            (r'^(yes|yeah|sure|okay|yep|definitely)\b', 'AGREEMENT / AFFIRMATION'),
            (r'^(no|nope|not really|never)\b', 'DISAGREEMENT / NEGATION'),
            (r'^(can you|could you|please|would you)\b', 'REQUEST'),
            (r'^(thanks|thank you|appreciate it)\b', 'THANKING'),
            (r'\?$', 'QUESTION'),
            (r'^(i think|it is|we have|the system)\b', 'STATEMENT / INFORM')
        ]

    def classify(self, utterance):
        text = utterance.strip().lower()
        for pattern, act in self.rules:
            if re.search(pattern, text):
                return act
        return 'INFORM / NEUTRAL'

classifier = DialogActClassifier()

conversation = [
    ("User", "Hello! Good morning."),
    ("Agent", "Hello! How can I assist you today?"),
    ("User", "Can you please check my account balance?"),
    ("Agent", "Sure, your current balance is $450."),
    ("User", "Okay, thank you so much!"),
```

```
    ("Agent", "You are very welcome. Goodbye!"),
    ("User", "Bye!")
]

print("--- Dialog Act Recognition Trace ---")
print(f"{'Speaker':<8} | {'Utterance':<38} | {'Recognized Dialog Act'}")
print("-" * 75)
for speaker, utt in conversation:
    act = classifier.classify(utt)
    print(f"{speaker:<8} | {utt:<38} | {act}")
```

OUTPUT

```
> python3 experiment_24.py
--------------------------------------------------------------------------------------------------------
--- Dialog Act Recognition Trace ---
Speaker  | Utterance                              | Recognized Dialog Act
---------------------------------------------------------------------------
User     | Hello! Good morning.                   | GREETING
Agent    | Hello! How can I assist you today?     | GREETING
User     | Can you please check my account balance? | REQUEST
Agent    | Sure, your current balance is $450.    | AGREEMENT / AFFIRMATION
User     | Okay, thank you so much!               | AGREEMENT / AFFIRMATION
Agent    | You are very welcome. Goodbye!         | INFORM / NEUTRAL
User     | Bye!                                   | FAREWELL
[Process completed with exit code 0]
```

RESULT

Thus, the dialog act recognition program successfully classified conversational dialogue utterances into pragmatic communicative acts.